Suggested topics for CV1 exam preparation

(Last update: 9 Oct 2026)
We have put together a checklist of core topics. The exam will test material from lectures, seminars, and computer labs. If a topic is covered in the computer lab assignments, there may be coding-related questions (see the Syllabus for the explanation of coding-related questions). **Important: While this list covers the core topics of the exam, they are not exhaustive and are here just to help you prioritize as a suggestion.**

Focus on both conceptual and mathematical understanding:

- Intrinsic matrix and extrinsic matrix (What are the elements? How to construct them?)
- Camera projection model (What does the world frame and camera frame mean? Where is the pinhole? How are images formed using this model? What is the homogeneous coordinate system and why use it?)
- Photometric stereo and reflectance (What are the necessary components to determine image pixel intensity? What do these components mean, and how to model them? How to estimate the surface normal and the depth?)
- Neighborhood processing (What are dilation, erosion, opening, and closing? How to perform these operations? How to do connected component analysis?)
- Image filtering and local features (What does convolution mean? How to do convolution? What does image gradients mean? What is a Canny edge detector?)
- Fourier Transform (What is Fourier Transform doing? Why does it need to use the complex plane with real and imaginary numbers? How do the real and imaginary numbers relate to the rotation angle, angular speed, magnitude, and phase?)
- RANSAC and keypoint matching (What is RANSAC? How to do keypoint matching?)
- Optical flow (What is optical flow? How to estimate optical flow using Lucas-Kanade?)
- Global Transforms and Image Stitching (How to do forward and backward image warping? What is homography and how to estimate it? How to do affine and projective transformations? What is a SIFT descriptor and how to build it?)
- Image classification (What are the metrics for evaluating image classification?)
- Retrieval, Detection, and Segmentation (How to do image retrieval using keypoints?)
- Convolutional Neural Networks and Object Detection (What is CNN and what elements are in the architecture? What is convolution operation and how to calculate it? How many training parameters in different types of layers?)

Prioritize conceptual understanding:

- Clustering (What is mean-shift? What is k-means?)
- Image classification (What is k-nearest neighbor? What is Bag-of-Visual-Words?)
- Retrieval, Detection, and Segmentation (What is Non-Maximum Suppression? What is Selective Search for object detection? What is “average precision”? What is “mean average precision”?)
- Convolutional Neural Networks and Object Detection (How do different optimizers, learning rates, and weight initialization strategies affect model performance? What are vanishing/exploding gradients and how to address the problems? What are R-CNN, Fast R-CNN, Faster R-CNN, and how are they different from each other?)
- Single Shot Detection and Other Network Architectures (What is Region Proposal Network? What is YOLO? What is ResNet?)
- Shape from X, Stereo, Structure from Motion (What is epipolar geometry?)